

Coronary artery reactivity in human vessels: some questions and some answers^{1,2}

STANLEY KALSNER

Department of Pharmacology, University of Ottawa Faculty of Health Sciences, Ottawa, Ontario, Canada K1H 8M5

The large surface arteries of the heart are of special interest in the study of cardiovascular dynamics and pathology. These so-called epicardial vessels are distinct from the intramyocardial or intramural vessels that penetrate into the heart muscle. In most organ systems, the smaller vessels within the organ substance are of most interest because they provide the resistance to blood flow, are well innervated by sympathetic nerves, and dilate in response to metabolic activity. The epicardial arteries of the heart, however, deserve careful attention for several important reasons. They are the vessels that become atherosclerotic; they restrict flow briefly or persistently to induce angina, myocardial infarction, and probably sudden death; and as we know now, they are not simply passive conduits for the transfer of blood, but vessels with a remarkable capacity to contract and relax and this capacity may be abnormal in cardiac disease. It also appears that no one species has epicardial coronary arteries that can serve adequately as a model for the human vessels.

SPONTANEOUS TONE AND RHYTHM

There is no single consistent pattern of spontaneous activity observed uniformly in the isolated coronary arteries of all species. Coronary artery ring preparations of all species tested show some degree of spontaneous tone. Relative to a maximal contraction to a full agonist, individual species can be described as showing little tone (less than 10–20%), namely the dog and monkey, or substantial amounts of spontaneous tone (more than 20%), namely pigs, sheep, cattle, and humans. The cattle and human coronary arteries are conspicuous in showing spontaneous rhythmic activ-

ity in addition to tone. These contractions may consist of large amplitude, slow cycle (about 0.2/min), or low-amplitude fast-cycle rhythms (about 0.8/min) that in some vessel segments merge into a state of sustained tone (Fig. 1). These data indicate the presence of an efficient conducting system between muscle cells not usually associated with large arteries, and one able to involve a large percentage of contracting units, as pointed out previously (13).

NOREPINEPHRINE

In the generally understood concept of autonomic regulation of blood vessel tone, the sympathetic neurotransmitter norepinephrine (NE) is assumed to constrict arteries. This does not appear to be uniformly the case in the epicardial coronary arteries where distinct species differences in the direction and quantity

of the response are noted. For example, the coronary arteries of sheep and pig relax completely to NE over the broad concentration range of 6×10^{-9} – 2×10^{-5} M, although small contractions may be observed with the lowest few concentrations in the pig. The large epicardial arteries of the monkey only contract in response to NE whereas those of cattle consistently relax over the test range of 6×10^{-9} – 2×10^{-5} M (17). The coronary vessels of the dog can be broken down into two popula-

ABSTRACT

Spasm of a conduit coronary artery, converting it into a major resistance vessel impeding myocardial blood flow, may have severe short- or long-term effects on cardiac rhythm and systolic ejection of blood. It is now clear that human coronary arteries in vitro contract to acetylcholine but that relaxation is the only response observed in dog coronary vessels. Acetylcholine is as powerful a constrictor of human coronary arteries, in terms of tension induced, as 5-hydroxytryptamine (5-HT) or histamine and is a substantially more powerful constrictor than norepinephrine. Field stimulation of coronary artery strips caused a vasoconstriction that was partially antagonized by atropine (3.45×10^{-6} M). An enhanced reactivity of the epicardial arteries of cardiac and older patients to several agonists was also observed and appears to provide a background against which a number of vasoactive agents might induce spasm. Coronary tissue from cardiac patients also contains stores of 5-HT and histamine, and the histamine levels are substantially increased above the values in vessels from noncardiac patients. Coronary artery spasm or contraction probably can be initiated by diverse intrinsic and extrinsic influences, including autonomic discharge from either the parasympathetic or sympathetic nervous system or from histamine or 5-HT, and probably no one agent or entity is causative in all cases.—**Kalsner, S.** Coronary artery reactivity in human vessels: some questions and some answers. *Federation Proc.* 44: 321–325; 1985.

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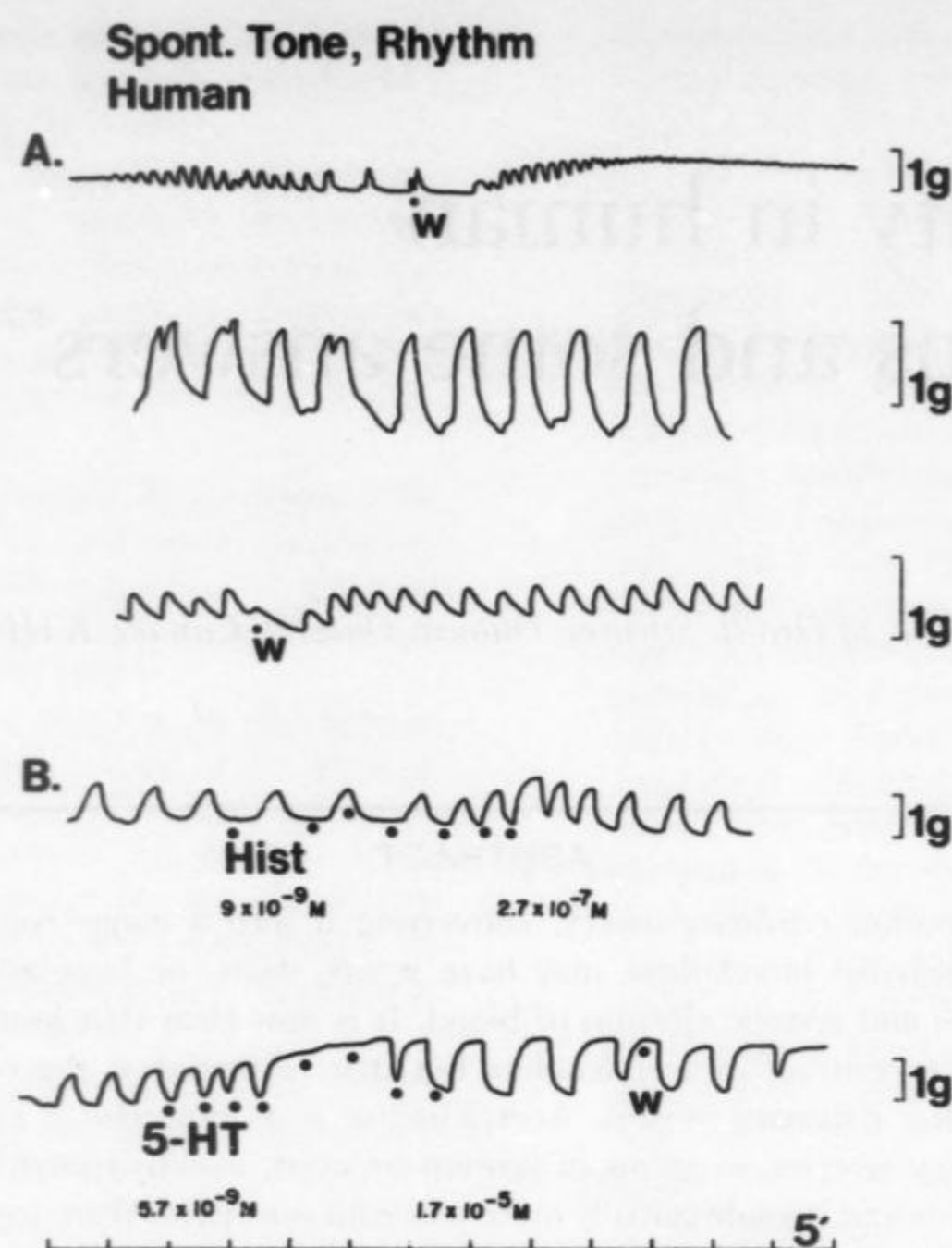


Figure 1. Spontaneous rhythm and tone in human coronary artery segments. A) Three common types of spontaneous contractions seen in coronary vessel segments. W indicates wash or change of bathing medium in muscle bath. B) Lack of effect of cumulative additions of histamine (9×10^{-9} – 2.7×10^{-7} M) (dots) and 5-HT (5.7×10^{-9} – 1.7×10^{-5} M) (dots) on tone in some vessel segments that show spontaneous activity.

tions: those that primarily contract to NE and those that mainly show relaxation. These contractile responses to NE in the dog vessels are of a considerably lesser magnitude than the responses of the same vessels to an agonist such as ergonovine.

Human coronary artery rings taken from the circumflex, right or left arteries respond to NE (6×10^{-9} – 2×10^{-5} M) with contractions. These large epicardial arteries do not relax to the catecholamine even though some of these vessels showed substantial degrees of spontaneous tone from which relaxation would have been readily visible. The concentration-response curve to NE obtained from 13 segments from 7 patients revealed that the maximal contraction to NE, however, was only 48.2% of that to 5-hydroxytryptamine (5-HT) in preparations taken from the same patients (Fig. 2).

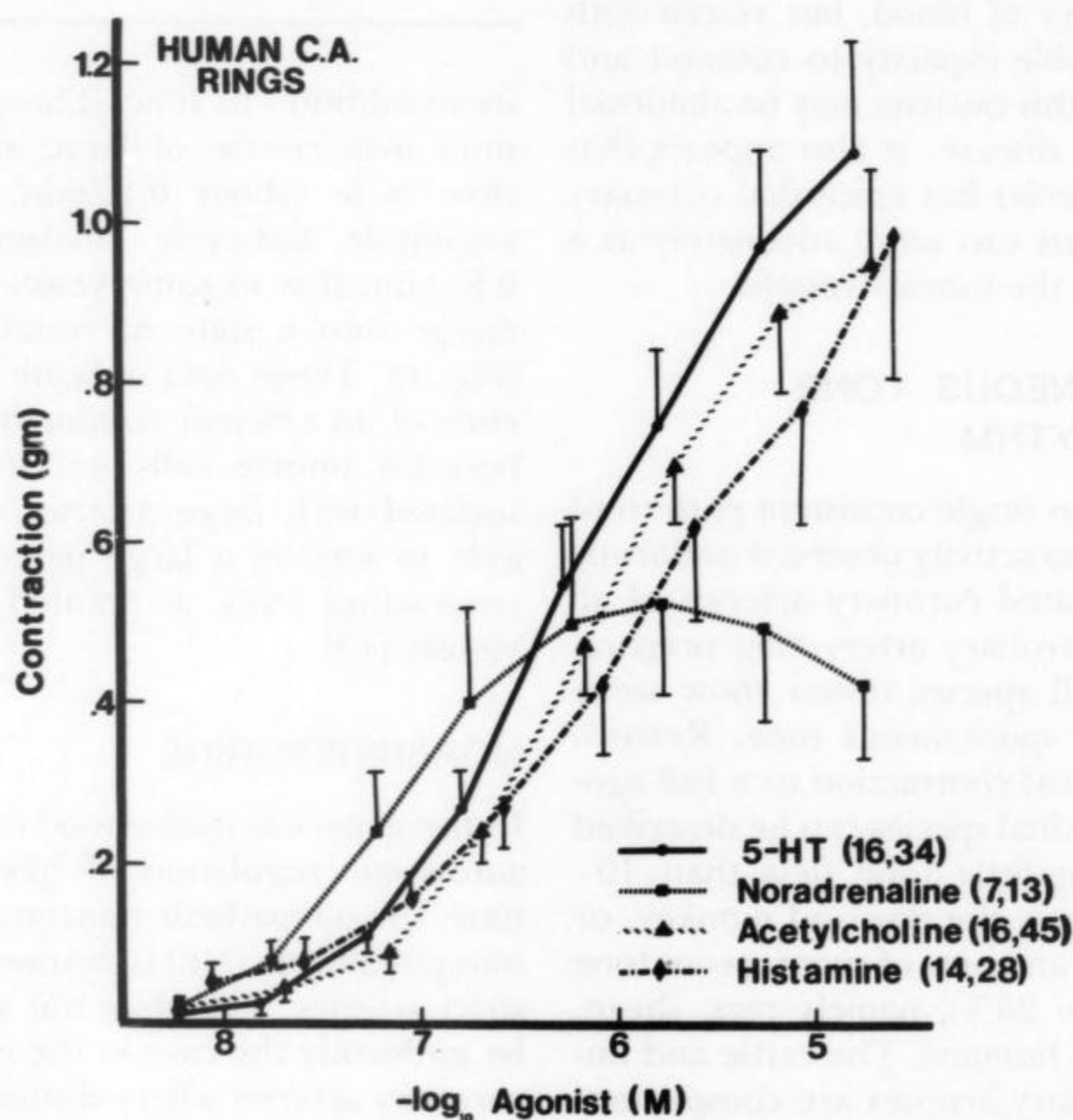
ACETYLCHOLINE

The transmitter of the parasympathetic nervous system, acetylcholine, has long been considered a dilator of human

coronary vessels, based mainly on extensive in vivo studies with the dog. The capacity of the ester to contract isolated coronary artery strips, observed occasionally, has generally been misinterpreted as an inexplicable peculiarity of large vessels maintained in an in vitro environment. However, it is now clear that this is not the case.

Isolated ring preparations of dog coronary vessels relax in response to acetylcholine (7×10^{-9} – 2×10^{-5} M) in vitro. Thus the dog vessels behave in vitro as they seem to in vivo. These in vitro relaxations are fully antagonized by denuding the intimal surface with the aid of a glass rod, which indicates that they are mediated by the endothelium. The dog, however, appears to be an exception in that the coronary vessels of other species tested (after removal from hearts within 30–60 min after death) only constrict to acetylcholine. Pig, sheep, and cattle epicardial vessels, maintained as ring preparations, respond to acetylcholine over the concentration range of 7×10^{-9} – 2×10^{-5} M only with contractions and these are not dependent on the presence of an intact endothelial surface, but appear to be mediated by muscarinic receptors on the smooth muscle cells.

Figure 2. Concentration-response curves of human coronary artery rings to cumulative concentrations of 5-HT, NE, histamine, or acetylcholine. Numbers in parentheses indicate number of patients whose vessels were sampled and the total number of segments tested.



Contractions of cattle coronary arteries to acetylcholine were studied in depth and found to be magnified considerably by inhibition of cholinesterase with eserine (1×10^{-6} M) (16). The concentration-response curve to acetylcholine showed that the cholinergic compound was as powerful a contractor as any tested on cattle vessels and considerably more potent than 5-HT.

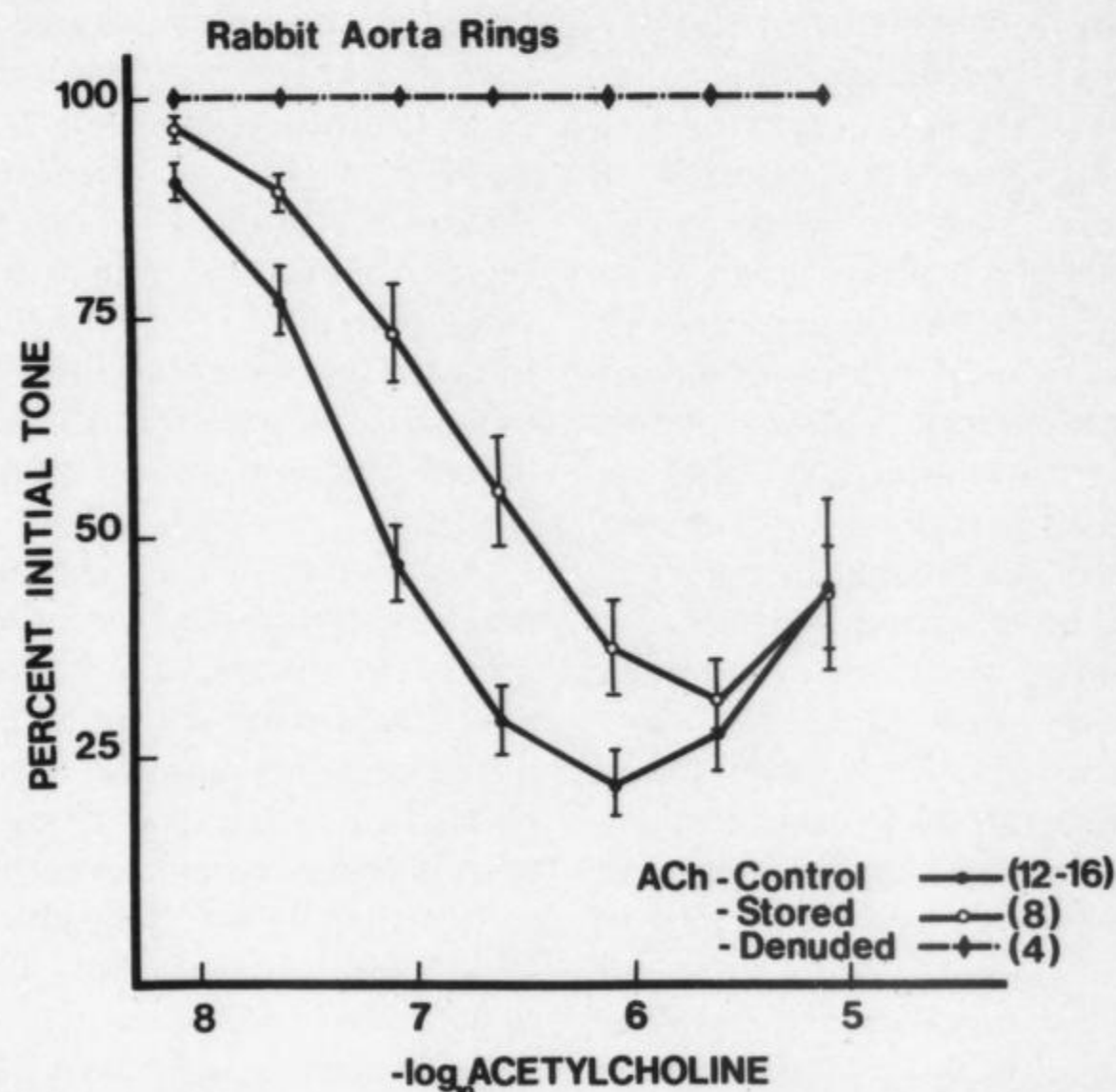
Seven epicardial artery preparations removed from the hearts of two monkeys within minutes after death were tested with acetylcholine; of these, two contracted and one relaxed partially. The monkey femoral artery, in contrast, showed a graded relaxation to increasing concentrations of acetylcholine that eliminated virtually all preexisting tone. Human coronary vessels either in the form of strips or ring preparations showed only contractions to acetylcholine. Relaxation was never observed. As shown in Fig. 2, 45 vessel segments (rings) from the epicardial coronary arteries of 16 patients contracted to acetylcholine and did so over the entire test concentration range.

Comparisons of the contractions produced by histamine, 5-HT, acetylcholine, and NE showed that whereas NE behaves as a partial agonist, acetylcholine is a full agonist producing contractions comparable in magnitude to those of 5-HT and histamine (Fig. 2).

The role of the endothelium in vascular relaxations to acetylcholine was examined with the rabbit aorta. As described by Furchgott (11), rings of aorta, or carefully prepared strips, relax to acetylcholine, and if the intimal surface is rubbed with a glass rod, this response is lost. To establish the durability of functional endothelium in cadaver tissue, rabbits were killed by cervical dislocation and portions of the aortas removed, mounted in muscle chambers as ring segments, and relaxations to acetylcholine confirmed (Fig. 3). These relaxations to acetylcholine (7×10^{-9} – 2×10^{-5} M) were obtained when tone was spontaneously generated or induced by 5-HT or NE. The remaining portion of the aortas was left in situ in the rabbits for 8 h, then removed and placed overnight in chilled, oxygenated Krebs solution. Only a very slight decrease in the magnitude of relaxation was noted (Fig. 3). These storage conditions were comparable to the most adverse encountered in my studies with human cadaver tissue.

The absence of relaxation to acetylcholine in human coronary artery rings is not caused by an inability of the vessels to relax. Prostacyclin (3×10^{-10} M– 3×10^{-6} M) completely relaxed vessels in a state of spontaneous tone or when tone was induced by agonists

Figure 3. Concentration-response curves of rabbit aortic rings to acetylcholine in control, stored and intinally denuded preparations. Number of preparations is shown in parentheses. See text for details.



Human Coronary Strips Field Stimulation

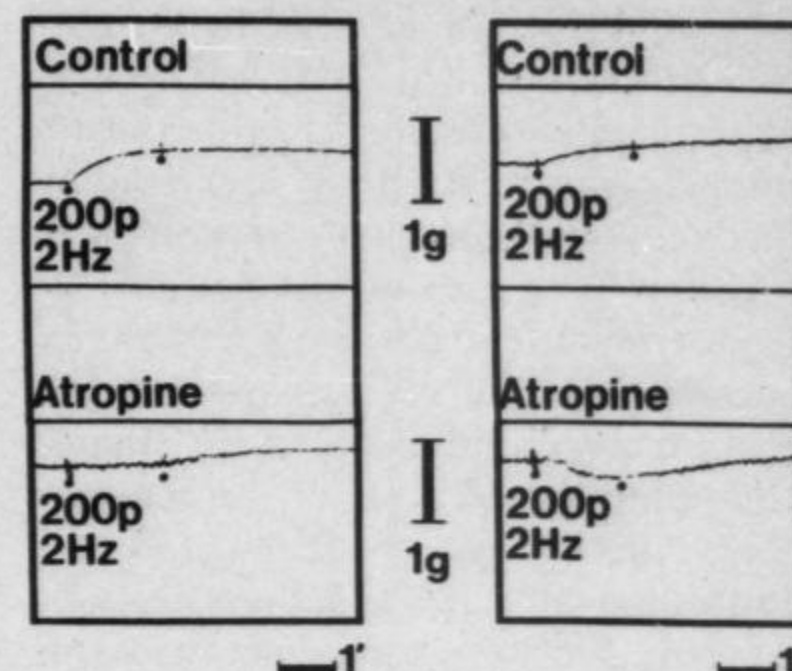


Figure 4. Responses of human coronary artery strips to field stimulation. Left and right responses of 2 strips to field stimulation with 200 pulses at 2 Hz in absence and presence of atropine (1.0×10^{-5} M).

such as acetylcholine. Additionally, the marked activity of acetylcholine as a vasoconstrictor does not seem to be a common property of all large arteries of humans. For example, pelvic branch and mesenteric arteries contracted in response to 5-HT and NE but not to acetylcholine (7×10^{-9} – 2×10^{-5} M).

Responses to nerve stimulation

Nonbeating slabs of ventricle were cut away from the hearts of cattle and perfused through the left anterior descending coronary artery by means of a Harvard peristaltic pump (16). Intraarterial infusions of acetylcholine (6.8×10^{-6} or 2.1×10^{-5} M) caused marked increases in perfusion pressure, which signified vasoconstriction, and these responses were increased substantially by prior inhibition of cholinesterase with eserine (1.1×10^{-6} M). Atropine (3.5×10^{-7} M) completely blocked these responses. Periarterial nerve stimulation at 2, 5, and 10 Hz induced vasoconstrictor responses that were also much increased by eserine and antagonized by atropine (16). The responses were not diminished, however, by hexamethonium (4.9×10^{-5} M). Pulse injections of NE (3 nmol) consistently relaxed the vascular preparation. No responses of the perfused coronary vessels attributable to sympathetic nerve stimulation were detectable.

When spirally cut strips of reactive human coronary artery were placed between parallel platinum electrodes and stimulated with 200 pulses at either

2 or 5 Hz, they contracted (Fig. 4). These responses were either partially or completely antagonized by atropine (3.5×10^{-6} or 1.0×10^{-5} M). Other experiments were done to rule out the unlikely possibility that acetylcholine mediates vasoconstrictor responses by a tyraminelike action to release NE from sympathetic nerves. For these experiments strips of human coronary arteries were preincubated in tritiated NE and mounted in superfusion assemblies. The tissues were stimulated with 250 pulses at 5 Hz and the effects on ^3H efflux of methacholine, a stable muscarinic analog of acetylcholine, were assessed. Increasing concentrations of the choline ester (6.2×10^{-8} – 6.2×10^{-5} M) progressively decreased the efflux of tritiated NE. The highest concentration of methacholine inhibited stimulation-induced efflux by 67%. This was an effect opposite to that required to explain the constrictor effects of acetylcholine.

Reports of the effects of parasympathetic nerve activation on the coronary circulation indicate that the response is negligible or that a dilation results (1, 6). However, the bulk of this work was done in the dog. The evidence that seems to confirm unequivocally a coronary vasodilator role for acetylcholine comes from the effects of injected acetylcholine, but again this is primarily from studies with the dog. These findings have then been generalized without verification to other species (2, 3, 10, 14, 15, 19, 20).

It appears that findings of coronary vessel constriction to acetylcholine have not been integrated into our theoretical constructs of autonomic-coronary vascular function, primarily because of the overwhelming volume of the studies performed on the dog. It is reinforced in the present work that the data collected with the dog appear to be fully accurate and reliable. The dog is exceptional, however, in its coronary vessel response to acetylcholine, yielding consistently only vasodilation whether tested in vivo or in vitro, and the species is consequently an inappropriate model from which to assess autonomic control of human coronary vessels.

Most other species, and in particular humans, show coronary vasoconstrictor responses even to low concentrations of the choline ester. This is not phylogenetically surprising. As pointed out by Burnstock (4), acetylcholine is mainly a constrictor of large elastic arteries in

fish and reptiles. It appears that such a function is maintained in the coronary arteries of most mammals. The constrictor response to acetylcholine appears to be a true atropine-sensitive, smooth muscle, mediated effect that is in contrast to the relaxation mediated by a presently unidentified factor released by the endothelium. Thus, the dog coronaries seem to lack significant functional smooth muscle receptors for acetylcholine, and have only the endothelial relaxing factor to mediate a response.

Autonomic innervation

We can conclude several relevant points about the innervation of the epicardial coronary arteries: 1) the vessels are innervated by both sympathetic and parasympathetic nerves and neither branch penetrates more than one-third into the media from the adventitial surface (7, 12); 2) the endothelium is on the wrong surface of the coronary artery to influence or mediate neurally evoked responses to acetylcholine, and 3) there is no evidence to support the notion that acetylcholine-mediated relaxation occurs at all in human coronaries. The function of endothelial-mediated vascular relaxation to acetylcholine, in those species and tissues where it occurs, is ambiguous.

Spasm and acetylcholine

The present work revealed that acetylcholine is a more potent coronary vasoconstrictor than NE in humans and is as powerful a contractor of the coronaries as 5-HT or histamine (Fig. 2). The amount of tension generated to a maximally effective concentration of acetylcholine is about two times as great as that to NE, which probably indicates that cholinergic rather than adrenergic mechanisms are most relevant to coronary flow regulation. This surprising finding makes it imperative that the participation of acetylcholine and cholinergic neurotransmission in coronary artery spasm be explored further.

The possibility of a primary coronary vasoconstrictor effect of acetylcholine and of parasympathetic stimulation should be integrated into our conception of coronary vascular dynamics and autonomic nerve control if we are to deal with the clinical implications of parasympathetic imbalance in humans. Acetylcholine, locally released in the coronaries, might be critically involved

in the etiology of spasm and sudden myocardial events, including sudden death (17). Such a selective liberation of acetylcholine could not be readily duplicated by the generalized administration of the substance to the circulation or even to the heart.

A few clinical reports suggest that cholinergic compounds can induce spasm in subsets of susceptible patients. For example, Yasue et al. (22) and Endo (8) have claimed that anginal episodes could be induced in such patients by the choline ester methacholine, a muscarinic analog of acetylcholine, or by pilocarpine. Interestingly, Engel (9) has pointed out in his psychiatric assessment of mental and emotional states at sudden death that "giving up or helplessness or hopelessness" (perhaps a parasympathetic state) is as common as "overwhelming excitation" in cases of sudden death.

To explain the involvement in spasm of cholinergic compounds, a class of presumed vasodilators, Yasue et al. (21) proposed that acetylcholine causes increased release of NE from sympathetic nerves "which in turn causes severe vasoconstriction of large coronary arteries" by α -receptor activation. As shown here such a hypothesis is untenable. The potent cholinergic agent methacholine decreased, and did not increase sympathetic transmitter release in humans or dogs.

Spontaneous rhythm and agonists

Human coronary artery rings that show active rhythmic contractions may still be essentially unresponsive to agonists such as 5-HT or histamine over a broad concentration range of 6×10^{-9} – 3×10^{-5} M (Fig. 1). Agonists can also initiate rhythm in some quiescent vessel segments but not induce further contractions (Fig. 1). These observations indicate fundamental differences in the mechanisms involved in receptor-mediated and intrinsically regulated contractions.

An analysis of the incidence of spontaneous rhythm revealed that ring segments were substantially more likely to manifest rhythm if they came from the hearts of older patients. The incidence of rhythm was three times greater in vessels from patients who were above rather than below the age of 55 years ($P < 0.005$). Rhythm was present in 47 out of 187 segments from 24 older patients and in only 7 of 79 segments from 12 younger patients.

Cardiac and noncardiac differences

The incidence of rhythm was significantly greater ($P < 0.025$) in the vessel segments taken from the hearts of patients whose death was attributed by the pathologist to cardiac disease and showed myocardial evidence of old or recent infarcts. Rhythm was present in 26 of 81 segments from 10 cardiac patients and 27 of 165 segments from 23 noncardiac patients. Experiments were done specifically to determine if coronary artery segments from cardiac and noncardiac patients respond differently to a variety of vasoactive substances (18). Whereas concentration-response curves to acetylcholine did not differ significantly in the vessel segments of patients from either category, the responses to histamine, 5-HT, and NE over the entire concentration-response range were substantially greater in the vessels obtained from cardiac patients (18). The maximal responses to these three agonists were significantly and substantially greater in the cardiac group.

Amine content

The coronary arteries contained surprisingly high amounts of 5-HT and histamine but a low concentration of catecholamines, the latter undoubtedly reflecting a relatively poor sympathetic innervation (18). No significant differences in the total concentration of NE and epinephrine (E) were detected between cardiac and noncardiac patients, but the levels of serotonin were slightly reduced and that of histamine greatly increased in the vessels of the cardiac group of patients (18).

Implications

The contractile status of the large surface arteries of the heart is important in the definition and therapy of several major forms of heart disease (17). Spasm of a conduit coronary artery, converting it into a major resistance vessel impeding myocardial blood flow, may have disastrous consequences for cardiac function and hence for the individual. The enhanced reactivity of the epicardial arteries of cardiac and older pa-

tients to several vasoactive substances appears to provide a background against which a number of vasoactive agents might induce spasm.

The finding that coronary tissue from cardiac patient contains stores of 5-HT and histamine, and that the latter are substantially increased above values in noncardiac patients, suggests that its sudden release, caused by injury or an allergic response as in an antigen-antibody type reaction (5), could induce a powerful contraction or spasm of a coronary vessel segment and precipitate a cardiac crisis.

Spasm probably can be initiated by diverse influences, including autonomic discharge from either the parasympathetic or sympathetic nervous system or from histamine or 5-HT, and probably no one agent or entity is causative in all cases. The in vitro study of large coronary vessels is sure to be a rich and fruitful area for further study, and with reasonable hope, it may provide us with the solution to a major facet of cardiac disease. 

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